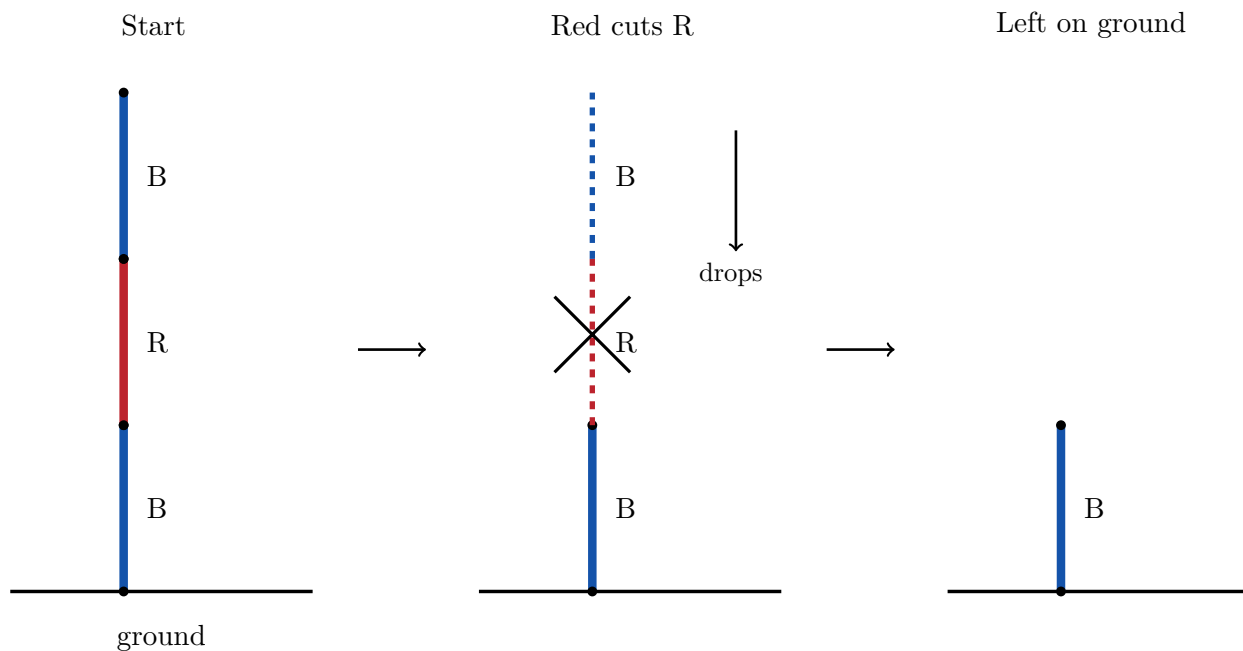


Blue removes one blue (B) segment. Red removes one red (R) segment. All segments cut off from the ground disappear. Take turns; change one stalk per turn. If you cannot move, you lose.



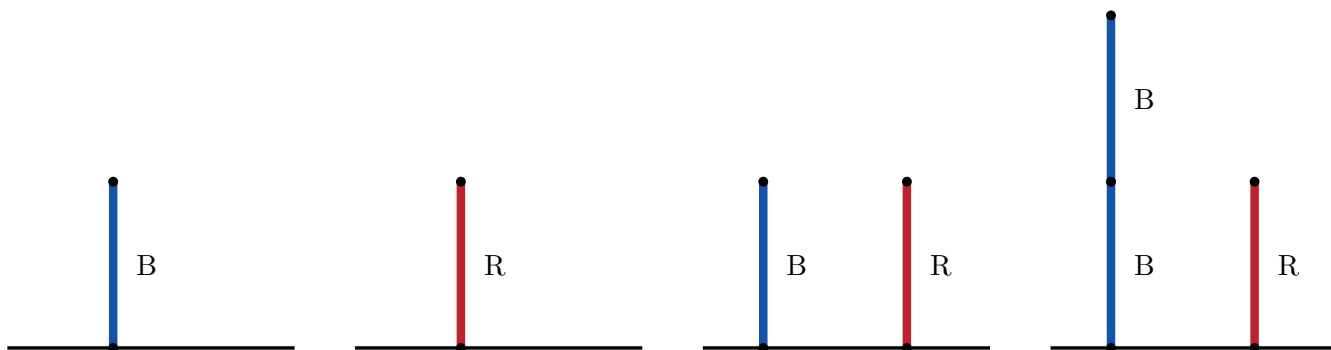
Problem 1: Play each board with Blue starting, then with Red starting. On which boards can one color always win?

A

B

C

D

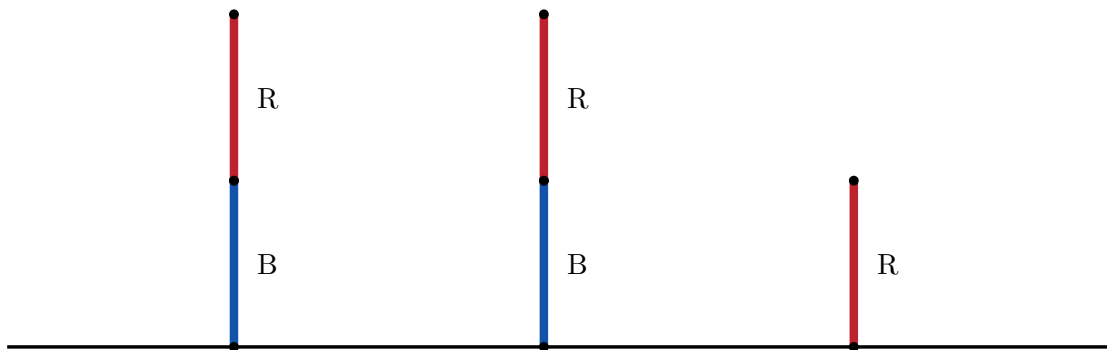


Problem 2: A board balances when the second player can always win, whichever color starts. Which of these boards balance?

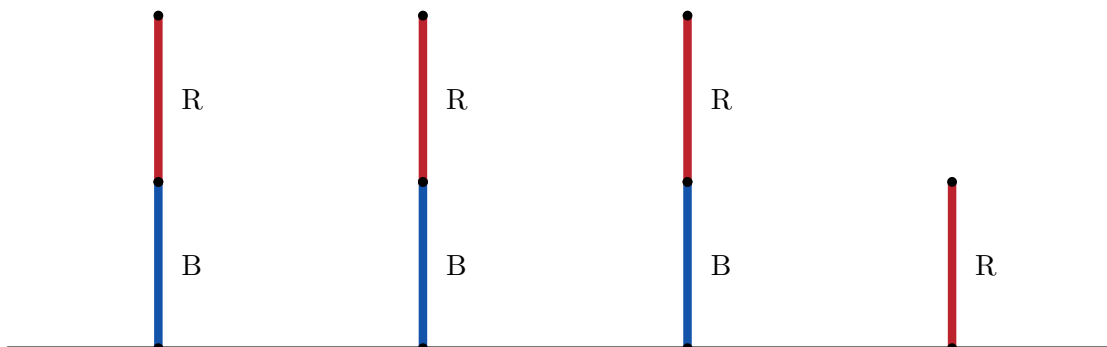
A



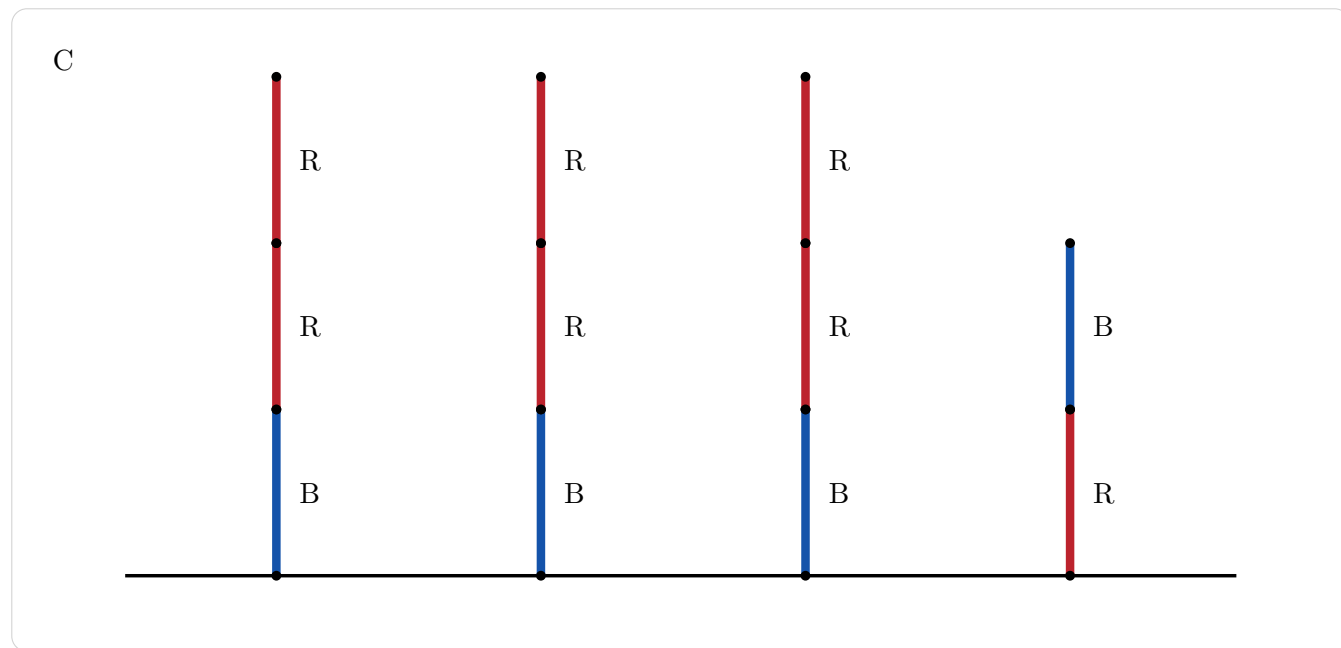
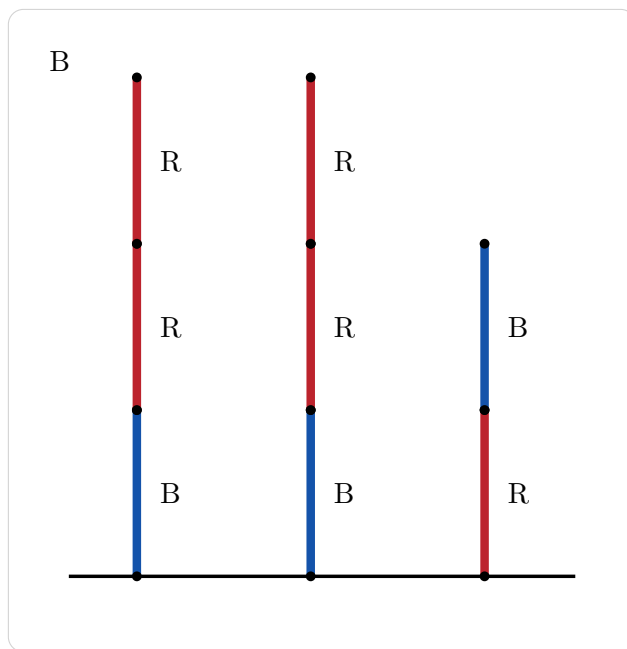
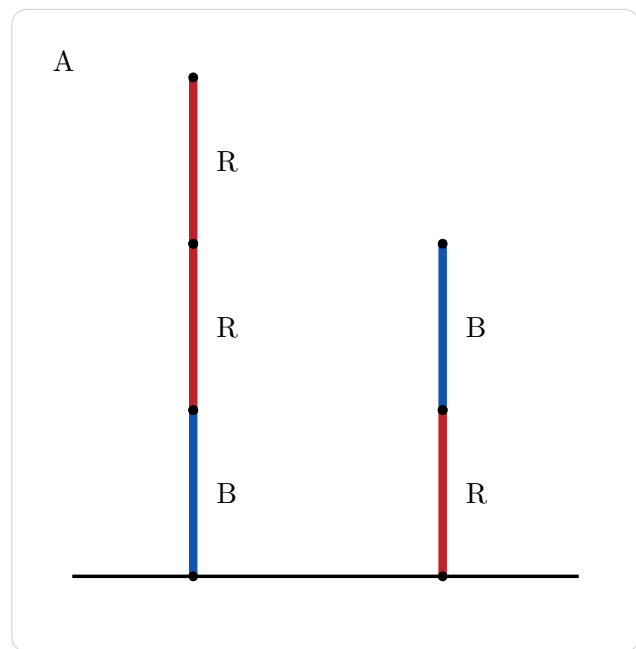
B



C



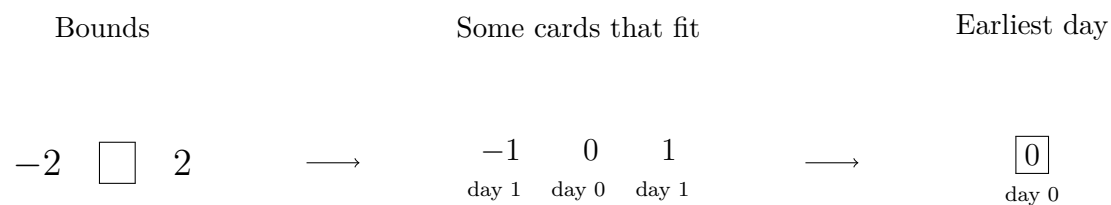
Problem 3: Which of these boards balance? Find a way for the winning color to win each other board, whoever starts.



Problem 4: Make two different boards that balance. Each board must include a stalk with at least two segments.



A number must be greater than every card on its left and less than every card on its right.
Choose the card from the earliest day. You may reuse cards.



Problem 5: Choose the oldest number card that fits each pair of bounds. Put it in the middle.

greater than 0		less than 3
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greater than −1		less than 1
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greater than 0		less than 1
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greater than $\frac{1}{2}$		less than 1
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Problem 6: Make two different pairs of bounds that choose the same number card. Make a pair that chooses a card newer than both bounds.

0	-1	1	-2
day 0	day 1	day 1	day 2

$-\frac{1}{2}$	$\frac{1}{2}$	2	-3
day 2	day 2	day 2	day 3

$-\frac{3}{2}$	$-\frac{3}{4}$	$-\frac{1}{4}$	$\frac{1}{4}$
day 3	day 3	day 3	day 3

$\frac{3}{4}$	$\frac{3}{2}$	3
day 3	day 3	day 3

Extra workspace

A large rectangular workspace for extra work, with a horizontal line near the bottom.